Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

Class:	**TE-AI&DS**	**Semester:**	**V**
Course Code:	**2015111**	**Course Name:**	**Statistics for Machine Learning and Data Science Lab**

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Roll No. :	64
Experiment No.:	**3**
Title of the Experiment:	**Implement Correlation Analysis, Distance Measures, and Data Preprocessing Techniques**
Date of Performance:	21/07/26
Date of Submission:	28/07/26

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :

Signature :

Date :

Aim: To implement correlation analysis, distance measures, and basic data preprocessing techniques on the Pima Indians Diabetes Dataset to identify relationships between variables and prepare data for further analysis.

Objectives

After completing this experiment, the student will be able to:

1. Apply correlation and distance measures to analyze relationships and similarities among data observations.
2. Implement basic data preprocessing techniques to prepare a dataset for statistical analysis and machine learning.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains medical diagnostic information for **768 female patients** and is used to study factors associated with diabetes.

The dataset contains numerical attributes and a binary outcome variable indicating whether the patient is diabetic.

Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Identify missing or invalid values, particularly zero values in medical attributes where zero may be unrealistic.
3. Handle selected missing values using an appropriate imputation technique.
4. Calculate the correlation matrix for numerical variables.
5. Visualize correlations using a heatmap.
6. Calculate Euclidean, Manhattan, and Cosine distances for selected observations.
7. Apply an appropriate normalization or standardization technique.
8. Compare the data before and after preprocessing.
9. Interpret the obtained correlation and distance measures.

Description of the Experiment

Correlation analysis helps determine the strength and direction of relationships between variables.

For example, the relationship between:

$$GlucoseandBMI$$

can be analyzed using correlation.

Distance measures determine the similarity or dissimilarity between observations.

Data preprocessing is performed to improve the quality and consistency of data before further statistical or machine learning analysis.

The experiment follows: Raw Dataset, Identify Invalid/Missing Values, Data Preprocessing, Correlation Analysis, Distance Calculation and Interpretation

Detailed Description of Statistical Methods

Correlation

Correlation measures the strength and direction of the linear relationship between two variables.

The Pearson correlation coefficient is:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

The value of $r$lies between:

$$-1 \leq r \leq 1$$

$r \approx 1$: Strong positive relationship

$r \approx -1$: Strong negative relationship

$r \approx 0$: Weak or no linear relationship

A correlation heatmap can be used to visualize relationships among multiple variables.

Euclidean Distance

Euclidean distance between two observations is:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

It represents the straight-line distance between two points.

Manhattan Distance

Manhattan distance is:

$$d(x, y) = \sum_{i=1}^{n} |x_i - y_i|$$

It measures distance by summing the absolute differences between corresponding features.

Cosine Distance

Cosine similarity is:

$$cos(\theta) = \frac{x \cdot y}{\|x\| \|y\|}$$

Cosine distance is:

$$d_{cosine} = 1 - cos(\theta)$$

It measures the difference in the direction of two vectors.

Data Preprocessing

Missing Value Handling

Invalid or missing values may be replaced using techniques such as:

a. Mean imputation

b. Median imputation

For medical variables, median imputation may be preferred when extreme values are present.

Normalization

Min-Max normalization transforms values into a specified range, usually [1]:

$$x^{'} = \frac{x - min(x)}{max(x) - min(x)}$$

Standardization

Standardization transforms a variable to have mean 0 and standard deviation 1:

$$z = \frac{x - \mu}{\sigma}$$

Scaling is important when distance-based methods are used because variables with larger numerical ranges can otherwise dominate the distance calculation.

Implementation:

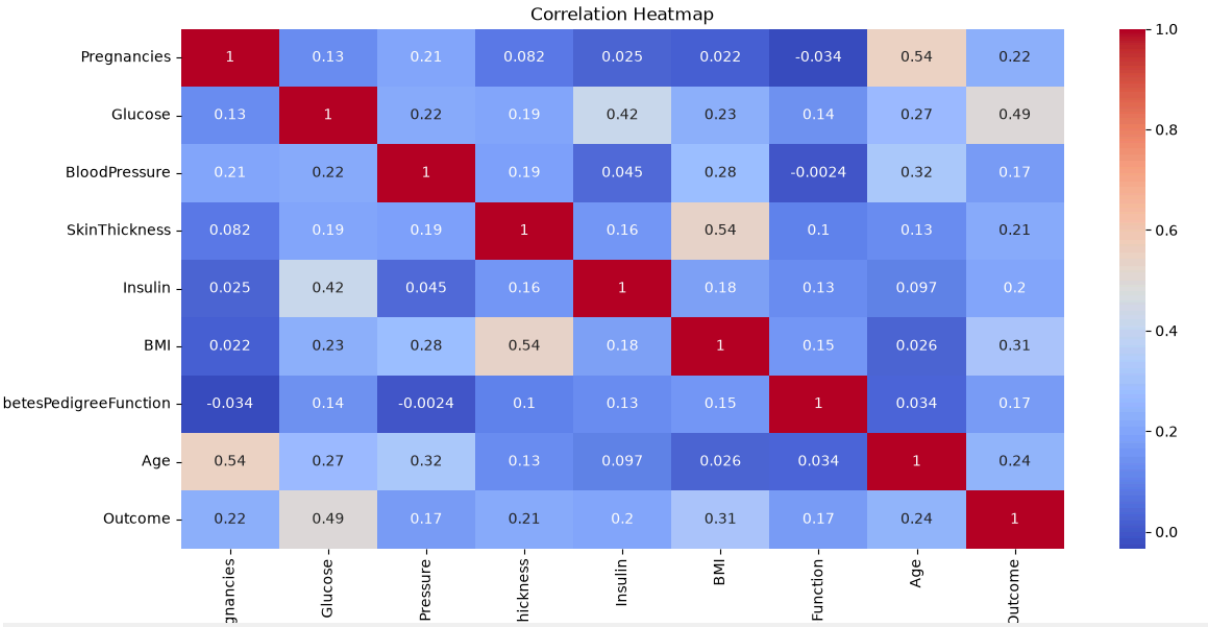
```
# Import Libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics.pairwise import euclidean_distances, manhattan_distances, cosine_distances

# 1. Load Dataset
data = pd.read_csv("diabetes.csv")
print("First 5 rows of Dataset:")
print(data.head())

# 2. Identify Missing/Invalid Values
print("\nZero values before preprocessing:")
columns = ["Glucose", "BloodPressure", "SkinThickness", "Insulin", "BMI"]
for col in columns:
    print(col, ":", (data[col] == 0).sum())

# 3. Handle Missing Values
# Replace zero values with NaN
data[columns] = data[columns].replace(0, np.nan)
# Median Imputation
imputer = SimpleImputer(strategy="median")
```

```
data[columns] = imputer.fit_transform(data[columns])
print("\nMissing values after imputation:")
print(data.isnull().sum())
# 4. Correlation Matrix
correlation = data.corr()
print("\nCorrelation Matrix:")
print(correlation)
# 5. Correlation Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(correlation,
            annot=True,
            cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
# 6. Distance Measures
# Selecting first two observations
x = data.iloc[[0]]
y = data.iloc[[1]]
euclidean = euclidean_distances(x,y)
manhattan = manhattan_distances(x,y)
cosine = cosine_distances(x,y)
print("\nDistance Measures:")
print("Euclidean Distance:",
      euclidean[0][0])
print("Manhattan Distance:",
      manhattan[0][0])
print("Cosine Distance:",
      cosine[0][0])
# 7. Normalization
scaler = MinMaxScaler()
normalized_data = scaler.fit_transform(data)
normalized_df = pd.DataFrame(normalized_data,
                             columns=data.columns)
# 8. Compare Before and After Preprocessing
print("\nOriginal Data:")
print(data.head())
print("\nNormalized Data:")
print(normalized_df.head())
```




```
PS D:\Statistics Lab AITE\Experiment 1> python exp3.py
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age  Outcome
0            6      148             72             35        0  33.6                     0.627   50        1
1            1       85             66             29        0  26.6                     0.351   31        0
2            8      183             64              0        0  23.3                     0.672   32        1
3            1       89             66             23       94  28.1                     0.167   21        0
4            0      137             40             35      168  43.1                     2.288   33        1
                          Pregnancies   Glucose  BloodPressure  SkinThickness   Insulin       BMI  DiabetesPedigreeFunction       Age   Outcome
Pregnancies                  1.000000  0.128213       0.208615       0.081770  0.025047  0.021559                 -0.033523  0.544341  0.221898
Glucose                      0.128213  1.000000       0.218937       0.192615  0.419451  0.231049                  0.137327  0.266909  0.492782
BloodPressure                0.208615  0.218937       1.000000       0.191892  0.045363  0.281257                 -0.002378  0.324915  0.165723
SkinThickness                0.081770  0.192615       0.191892       1.000000  0.155610  0.543205                  0.102188  0.126107  0.214873
Insulin                      0.025047  0.419451       0.045363       0.155610  1.000000  0.180241                  0.126503  0.097101  0.203790
BMI                          0.021559  0.231049       0.281257       0.543205  0.180241  1.000000                  0.153438  0.025597  0.312038
DiabetesPedigreeFunction    -0.033523  0.137327      -0.002378       0.102188  0.126503  0.153438                  1.000000  0.033561  0.173844
Age                          0.544341  0.266909       0.324915       0.126107  0.097101  0.025597                  0.033561  1.000000  0.238356
Outcome                      0.221898  0.492782       0.165723       0.214873  0.203790  0.312038                  0.173844  0.238356  1.000000
```

Conclusion

Correlation analysis, distance measures, and data preprocessing techniques were implemented on the Pima Indians Diabetes Dataset. Correlation analysis helped identify relationships between medical variables, while distance measures quantified the similarity or dissimilarity between observations. Data preprocessing techniques helped improve the consistency and suitability of the data for further analysis and machine learning applications.

Questions to be Answered by Students

1. Which pair of variables has the strongest positive correlation in the dataset?

   Strongest Positive: Pregnancies & Age and **SkinThickness & BMI** (tied at **0.54**).

   Strongest Negative: Pregnancies & DiabetesPedigreeFunction (**-0.034**), though effectively zero/negligible.

2. Which pair of variables has the strongest negative correlation, if any?

   Euclidean: Straight-line geometric distance.

   Manhattan: Grid-based path distance (sum of absolute differences).

   Cosine: Directional/angle difference between feature vectors, ignoring absolute scale.

3. How do the Euclidean, Manhattan, and Cosine distances differ for the selected observations?

   Technique: Imputation (replacing invalid 0 values with the median, mean, or KNN estimates).
   Why: Avoids discarding data (dropping rows) while preventing zero values from distorting distance and statistical calculations.

4. Which preprocessing technique was applied to the invalid or missing values, and why was it selected?

   Without Scaling: High-magnitude variables (e.g., Insulin) dominate distance calculations.

   With Scaling (Min-Max / Z-score): Equalizes feature ranges so every variable contributes fairly to Euclidean and Manhattan distances.

5. How does normalization or standardization affect the calculated distance between observations?

   Without Scaling: High-value variables (e.g., *Insulin*) completely dominate distance calculations, making small-value variables (e.g., *Pedigree Function*) irrelevant.
   With Scaling: Every variable contributes equally to **Euclidean** and **Manhattan** distances by placing all features on a shared scale (e.g., 0 to 1 or standard deviation units).